

Jan-Setu – Civic Complaint Aggregator

SE Project | Option 2: Civic Services Platform

Product Overview: Jan-Setu lets citizens file civic complaints via a **web app** or **WhatsApp bot** after a one-time registration with their WhatsApp-linked mobile number. Complaints include location, a text or voice description, and an optional photo – processed by an **LLM** for classification, image verification, and department routing. Non-priority complaints are aggregated before forwarding; priority complaints are filed immediately.

User Identification

1.1 Primary Users – Citizens

Profile A – Low-Digital-Literacy Citizens

Representative: Pintu Yadav – farmer & shopkeeper, Bihar

- WhatsApp is primary digital tool; no awareness of online complaint portals
- Currently files complaints via multiple physical office visits; often pays informal fees (₹100-200)
- Comfortable with guided WhatsApp flows (validated via Aarogya Setu experience)

Profile B – Digitally Aware, Process-Ignorant Citizens

Representatives: Aditya Singh, Namrita Singh, Teena Khanna

- Comfortable with smartphones; unaware civic complaint portals exist
- Has attempted existing channels (toll-free numbers, office visits) – frustrated by zero tracking or follow-up

Profile C – Tech-Savvy Citizens

Representative: Harish Kumar – software engineer

- High digital fluency; willing to use a well-designed platform immediately
- Blocked by inability to identify which department handles which complaint type

1.2 Secondary Users – Government Officials/ Complaint Reciever

Profile D – Municipal Complaint Clerks

Representative: Sanjay Kumar – Head Clerk, Patna Nagar Nigam

- Handles 60–250 complaints/day; entire workflow runs on physical registers
- Incomplete complaints (missing address, contact, description) add 24–48 hrs to processing

Profile E – Utility Department Clerks

Representative: Savitur Tyagi – Clerk, Bijli Vibhag, West Midnapore

- Department portal exists but is severely underused; most complaints arrive in person
 - Frequently receives misdirected complaints outside department jurisdiction
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1.3 Tertiary Users – Indirect Stakeholders

Profile F – Field Officers

Representative: Sarvesh Pandey – CISF

- Receives complaints via manual forwarding chains; no direct intake from citizens
- Direct routing would enable faster response to urgent issues

Profile G – Institutional Administrative Staff

Representative: Vivek Kumar – Library Staff

- Mirrors civic complaint failures at institutional scale – manual workflows, no tracking, no automation
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User Research

1. Research Summary

We conducted 9 interviews across citizens and government officials – spanning a rural farmer in Bihar, a youth political activist, urban working professionals, a municipal head clerk, a utility department clerk, a CISF field officer, and institutional administrative staff. Across both sides of the civic complaint system, a consistent pattern emerged: the current process is entirely manual, relationship-dependent, and opaque. Citizens – regardless of education or digital literacy – are unaware that online complaint channels exist and default to physically visiting government offices, often multiple times, with no guarantee of resolution. On the government side, clerks are overwhelmed by high complaint volumes, incomplete submissions, and misdirected cases that waste field resources. Neither side has any real-time visibility into the complaint lifecycle – citizens don't know if their complaint was received, and officials have no automated tools to communicate back. The gap is not one of intent; both citizens and officials want the process to work. What's missing is a structured, accessible, and transparent channel that bridges them.

2. Pain Point

2.1 Citizens

	Pain Point
P1	Filing requires multiple physical visits; rural users face informal bribe pressure without personal connections
P2	Citizens cannot identify which department is responsible for their complaint
P3	No acknowledgment or tracking after submission – no visibility post-filing
P4	Online complaint portals exist but are entirely unknown to users across all profiles
P5	Government platforms use formal language; rural users lack the literacy to engage
P6	Relationship-based processes create access inequality – connected citizens get faster resolution

2.2 Government Officials/ Complaint Reciever

	Pain Point
P7	Complaints arrive missing critical fields (contact, landmark, description) – adds 24-48 hrs to processing
P8	High volume of misdirected complaints wastes staff time and field resources
P9	Classification and routing is fully manual; inter-department forwarding batched only twice daily
P10	No automated citizen communication – follow-up calls and repeat visits overload staff

3. SMART User Stories

3.1 – One-Time Registration via WhatsApp-Linked Mobile Number

As a citizen, **I want to** register on Jan-Setu using my WhatsApp-linked mobile number, **so that** I can access both the web app and WhatsApp bot under a single identity.

Point	Details
Specific	OTP sent to WhatsApp-linked mobile; one account grants access to both channels
Measurable	OTP delivered within 60 sec; registration completed under 2 min

Point	Details
Actionable	OTP verification via WhatsApp Business API
Relevant	Removes first-time digital entry barrier – P4, P6
Time-bound	Full registration in under 2 minutes

Acceptance Criteria:

- OTP delivered within 60 seconds; registration completes in under 2 minutes
- One account grants access to both web app and WhatsApp bot
- UI available in Hindi and English

3.2 – Voice Note Complaint Filing via WhatsApp Bot

As Pintu Yadav, **I want to** describe my complaint in Hindi via a WhatsApp voice note, **so that** I can file an issue without typing or visiting a government office.

Point	Details
Specific	Voice note transcribed by Sarvam.ai multilingual STT via the Jan-Setu WhatsApp bot
Measurable	Transcription returned within 30 sec; user confirms before submission
Actionable	Sarvam.ai STT API integrated into WhatsApp bot pipeline
Relevant	Removes language and literacy barriers – P1, P5
Time-bound	Transcription returned within 30 seconds of voice note submission

Acceptance Criteria:

- Bot accepts voice notes up to 2 minutes; transcription returned within 30 seconds
- Transcribed text shown to user for confirmation before filing
- Low-confidence transcriptions prompt the user to re-record or type

3.3 – Location-Based Routing to Correct Municipal Department

As Namrita Singh, **I want** Jan-Setu to automatically assign the right department based on my complaint and location, **so that** I don't need to know government structure to file correctly.

Point	Details
Specific	LLM classifies incident type; location maps complaint to the correct municipal jurisdiction

Point	Details
Measurable	Department assigned within 10 sec; shown to user before final submission
Actionable	LLM classification via OpenRouter + geolocation-to-department mapping
Relevant	Eliminates department-identification barrier — P2
Time-bound	Classification completed within 10 seconds of submission

Acceptance Criteria:

- Department auto-assigned within 10 seconds based on location and complaint content
- User can view and correct the department assignment before confirming
- Misrouted complaints flagged for manual clerk review within 1 hour

3.4 – LLM Image Verification Against Complaint Description

As Sanjay Kumar, **I want** the system to flag complaints where the photo doesn't match the text description, **so that** inconsistent submissions are caught before reaching my dashboard.

Point	Details
Specific	Multimodal LLM compares submitted image to complaint text and scores consistency
Measurable	Verification completed within 15 sec; flagged complaints visually marked in the dashboard
Actionable	Multimodal LLM image-text comparison via OpenRouter
Relevant	Filters unverified or misrepresented submissions — P7
Time-bound	Result available within 15 seconds of submission

Acceptance Criteria:

- Low-consistency complaints are flagged in the clerk's dashboard, not auto-rejected
- Verification result stored in the complaint audit trail

3.5 – Complaint Classification as Short-Term Fix or Long-Term Project

As Sanjay Kumar, **I want** complaints to arrive pre-tagged by resolution scope, **so that** I can prioritise work without manual triage.

Point	Details
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Point	Details
Specific	LLM classifies each complaint as "Short-Term Fix" or "Long-Term Project" at intake
Measurable	Every complaint arrives tagged; zero unclassified complaints reach the dashboard
Actionable	LLM prompt-based classification via OpenRouter
Relevant	Reduces manual classification overhead – P9
Time-bound	Classification completed within 10 seconds of submission

Acceptance Criteria:

- Every complaint is tagged before entering the clerk's queue
- Short-term and long-term complaints are displayed in separate dashboard sections
- Clerk can override the classification with a single action

3.6 – Aggregation Time Window for Non-Priority Complaints

As Sanjay Kumar, **I want** duplicate non-priority complaints about the same issue to be merged before submission, **so that** my department receives one consolidated report instead of dozens of individual duplicates.

Point	Details
Specific	Non-priority categories (sanitation, potholes, streetlamps, drainage) held in a location-based time window; duplicates merged before forwarding
Measurable	Department receives a single consolidated entry with affected-citizen count attached
Actionable	Location-clustering logic + configurable time-window rules in Jan-Setu backend
Relevant	Reduces duplicate complaint volume – P9, P10
Time-bound	Consolidated report forwarded at close of the configured time window

Acceptance Criteria:

- Complaints within the same location radius and category are merged during the window
- Consolidated report includes count of citizens who reported the same issue
- Filing citizens receive confirmation that their complaint is part of a consolidated submission

3.7 – Immediate Filing for Priority Complaints

As Aditya Singh, **I want** priority complaints (e.g., animal cruelty) to bypass the aggregation window and be filed immediately, **so that** urgent issues aren't delayed by routines designed for non-critical matters.

Point	Details
Specific	Priority complaint categories bypass the aggregation window; forwarded instantly upon LLM classification
Measurable	Complaint reaches the responsible department within 5 minutes of submission
Actionable	Routing rules override aggregation logic for defined priority categories
Relevant	Ensures urgent issues are not held alongside routine complaints – P1
Time-bound	Department receives complaint within 5 minutes of citizen submission

Acceptance Criteria:

- Priority complaints bypass aggregation and are forwarded within 5 minutes of classification
 - Citizen receives immediate acknowledgment confirming the complaint is fast-tracked
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3.8 – Complaint Status Tracking on Web App

As Teena Khanna, **I want to** view my complaint's live status on the Jan-Setu web app, **so that** I don't need to call or visit anyone for an update.

Point	Details
Specific	Web app displays complaint lifecycle: Drafted → Submitted → Processing → Completed
Measurable	Status updated within 24 hrs of any official action; user notified on every change
Actionable	Status-tracking backend synced with the Jan-Setu complaint pipeline
Relevant	Eliminates post-submission invisibility – P3
Time-bound	Status updated within 24 hours; 7-day inactivity triggers an escalation flag

Acceptance Criteria:

- Status updated within 24 hours of any department action
 - User receives a notification on every status change
 - No status change for 7 days triggers an automatic escalation flag
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3.9 – Status Request via Jan-Setu WhatsApp Bot

As Pintu Yadav, **I want to** check my complaint status by messaging the Jan-Setu bot in Hindi, **so that** I get an update without opening the web app.

Point	Details
Specific	WhatsApp bot returns current complaint status on user request
Measurable	Status returned within 30 seconds of the user's message
Actionable	WhatsApp Business API + backend status query from Jan-Setu database
Relevant	Extends status visibility to non-web-app users – P3
Time-bound	Response delivered within 30 seconds

Acceptance Criteria:

- Bot returns current complaint status within 30 seconds in plain Hindi
 - Resolved complaints include a brief summary of action taken
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3.10 – Mandatory Structured Intake on Both Channels

As Sanjay Kumar, **I want** all incoming complaints to arrive with validated mandatory fields, **so that** I can act on them the same day without chasing missing information.

Point	Details
Specific	Jan-Setu enforces name, mobile number, location, and issue description on web app and WhatsApp bot
Measurable	Zero complaints reach the dashboard with missing mandatory fields
Actionable	Form validation on web app; step-by-step guided flow on WhatsApp bot
Relevant	Eliminates the 24–48 hr delay from incomplete complaints – P7
Time-bound	Validation enforced at submission; incomplete complaints cannot be forwarded

Acceptance Criteria:

- Web app blocks submission for missing fields; user prompted in their language
 - WhatsApp bot does not advance without a valid response for each required field
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Interview Reference Link

G-Drive - https://drive.google.com/drive/folders/1KzqqPEAGwH1GPJJnpcSr4ZuV-n0tix-1?usp=drive_link